

Zero Day Guidelines

GENESIS
INNOVATE, IDEATE, INSPIRE

Welcome to Zero Day, a premier cybersecurity competition designed to test the analytical, offensive, and defensive skills of participating students. The event is structured to challenge participants in realistic cybersecurity scenarios.

1. Event Structure

Zero Day is divided into two distinct phases, designed to evaluate both puzzle-solving abilities and practical security operations:

- Phase 1: Capture The Flags (CTF)
- Phase 2: Red Team vs. Blue Team

2. Team Composition

Participation requires teamwork and collaboration. Each participating school must adhere to the following composition rule:

- A maximum of 2 students per team is allowed for this event.

3. Phase Details & Scoring

Phase 1: Capture The Flag (CTF)

In this phase, teams will act as investigators and ethical hackers to uncover hidden flags across various domains of cybersecurity.

- Categories: Web vulnerabilities, Cryptography, Steganography, and Open-Source Intelligence (OSINT) puzzles.
- Scoring: Each discovered flag will increment the team's points. The point value awarded is weighted according to the difficulty level of finding the respective flag.

Phase 2: Red Team vs. Blue Team

This phase simulates a live attack-and-defend scenario. Teams will be assigned to either the offensive (Red) or defensive (Blue) side.

- Team Assignment: Segregation into Red and Blue teams will be conducted on a strictly first-come, first-serve basis to ensure an equal distribution of schools in each team.

Role	Objective	Scoring Mechanism
Blue Team (Defend)	Will be provided with a website's source code. Their goal is to identify and patch zero-day vulnerabilities and bugs before they can be exploited.	Defense points are awarded for every flag <i>not</i> found by the Red Team (indicating a successfully patched vulnerability).
Red Team (Attack)	Will be provided with the target website's URL. Their goal is to find and exploit zero-day vulnerabilities within the infrastructure.	Attack points are awarded for every flag successfully found and submitted (indicating a successful vulnerability exploitation).

4. Authorized Tools and Restrictions

To ensure a fair and competitive environment, the following rules regarding tooling and scripts are strictly enforced:

4.1 Allowed Tools

- Standard Security Tools: Command-line utilities and network analysis tools such as curl, wget, nmap, wireshark, etc.
- Deciphering Tools: Specific software utilized for Steganography and Cryptography decoding.
- Artificial Intelligence: The use of Artificial Intelligence (AI) assistants and tools is explicitly allowed during the event.

4.2 Restricted Elements

- Pre-Written Exploit Scripts: The use of pre-written exploit payload scripts is strictly NOT allowed by default.
- Exception Process: If a team wishes to use a pre-written script, it must first be presented in plaintext to the event manager for approval.
- Deadline: Any requests to allow pre-written scripts must be submitted well in advance. Any requests for allowing pre-written scripts 24 hours (or less) before the event will be rejected.